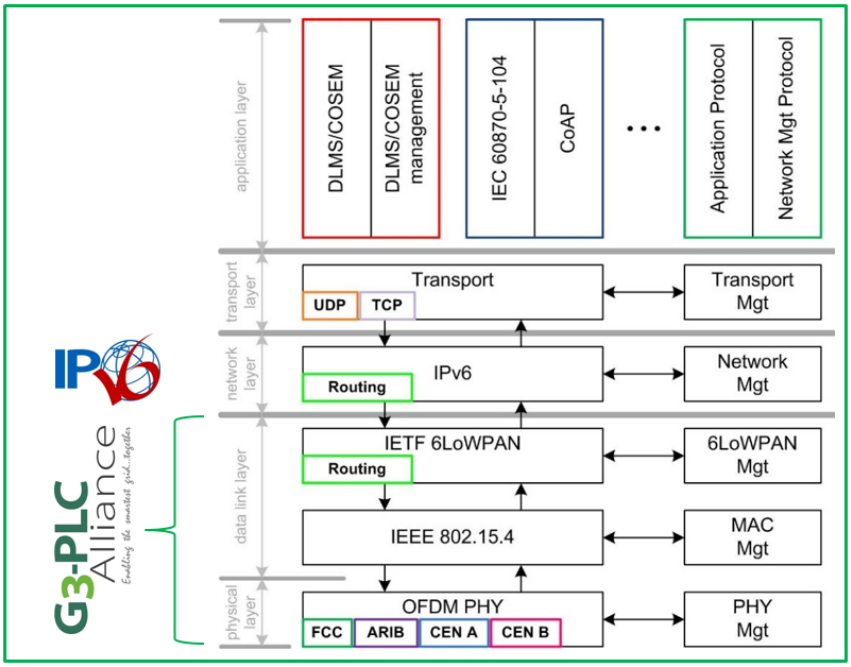
G3-PLC User Guidelines

Introduction of G3-PLC for non-experts (version 1.2, 03/2022)

Executive Summary

This paper is an introductory document to the G3-PLC technology and precedes a series of user guidelines published by the G3-PLC Alliance, intended to help end-users and manufacturers in implementing and deploying G3-PLC products

In this document, the G3-PLC protocol stack and its main characteristics will be briefly presented as well as an introduction to the G3-PLC Alliance and its certification program.



About G3-PLC

G3-PLC facilitates high-speed, highly-reliable, long-range communication over the existing powerline grid. The features and capabilities of G3-PLC have been developed to address the difficult challenges of powerline communications. While earlier approaches were a step in the right direction, they fall short of meeting the technical and reliability requirements necessary in the hostile environment of PLC.

G3-PLC meets these requirements because of its unique features such as a mesh routing protocol to determine the best path between remote network nodes, a “robust” mode to improve communication under noisy channel conditions and channel estimation to select the optimal modulation scheme between neighbouring nodes. Furthermore, its support of IPv6, enabling easy integration of various application profiles, adds high versatility and carries G3-PLC well into the future.

G3-PLC is an open, international standard published by ITU: https://www.itu.int/rec/T-REC-G.9903.

To help adopters properly integrate the G3-PLC protocol stack in products and systems we have developed a set of user guidelines. All user guidelines can be found on the G3-PLC Alliance website: http://www.g3-plc.com/what-is-g3-plc/userguidelines

For more information: www.g3-plc.com

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1. Overview of the G3-PLC technology

1.1. The G3-PLC protocol stack

G3-PLC [1] is a narrowband power line communication (NB-PLC) OFDM technology based on state-of-the-art narrowband PLC standards (the reference published international standard is ITU-T G.9903 [3] while power spectral density is defined in ITU-T G.9901 [2]) operating in several bandplans covering the overall frequency range 3 – 500 kHz to address numerous use cases matching regional regulation, as shown in Figure 1:

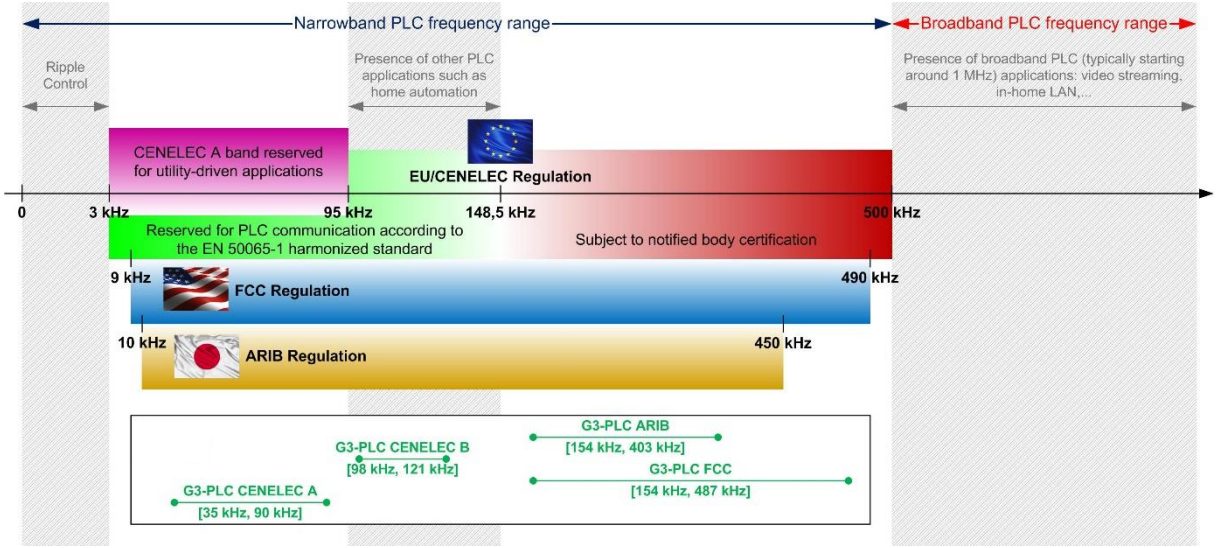


Figure 1: Different bandplans to address different needs and match regional regulation

The G3-PLC communication stack consists of a specific OFDM physical layer, an IEEE 802.15.4 [4] based MAC layer and an IETF 6LoWPAN-based adaptation layer allowing native support of IPv6. The ITU-T G.9903 [3] standard is describing the G3-PLC lower layers (physical and data link layers). The G3-PLC communication stack is maintained, promoted and certified by the G3-PLC Alliance.

The modular design of the stack allows the selection of different options at different layers (especially frequency bandplan and routing).

In addition, support of the IPv6 protocol [5] grants end-user flexibility to fulfil business requirements when choosing the appropriate higher layers (transport and application layers). This key feature also secures G3-PLC infrastructures in the long term, thanks to the scalability and future application compatibility provided by IPv6.

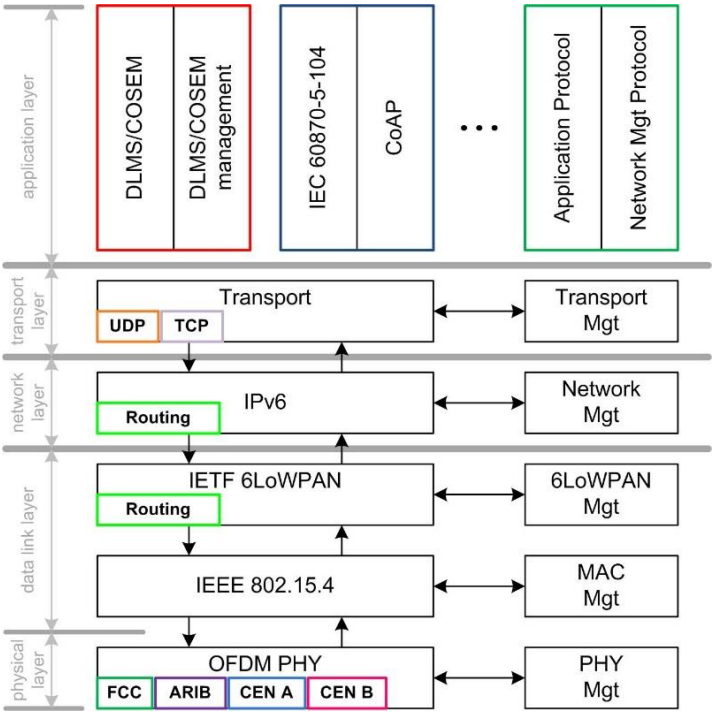


Figure 2: Typical G3-PLC communication stacks

1.2. G3-PLC main technical features in a nutshell

As pointed out previously, G3-PLC is an internationally standardized communication protocol stack gathering the advantages of an IPv6 compatible solution and offering high performances at the data link and physical layers.

Indeed, the OFDM physical layer embeds **powerful error correction** mechanisms based on a Reed-Solomon encoder followed by a convolutional encoder and a two-dimensional interleaver introducing time and frequency diversity, as well as an additional redundancy scheme with repetition codes.

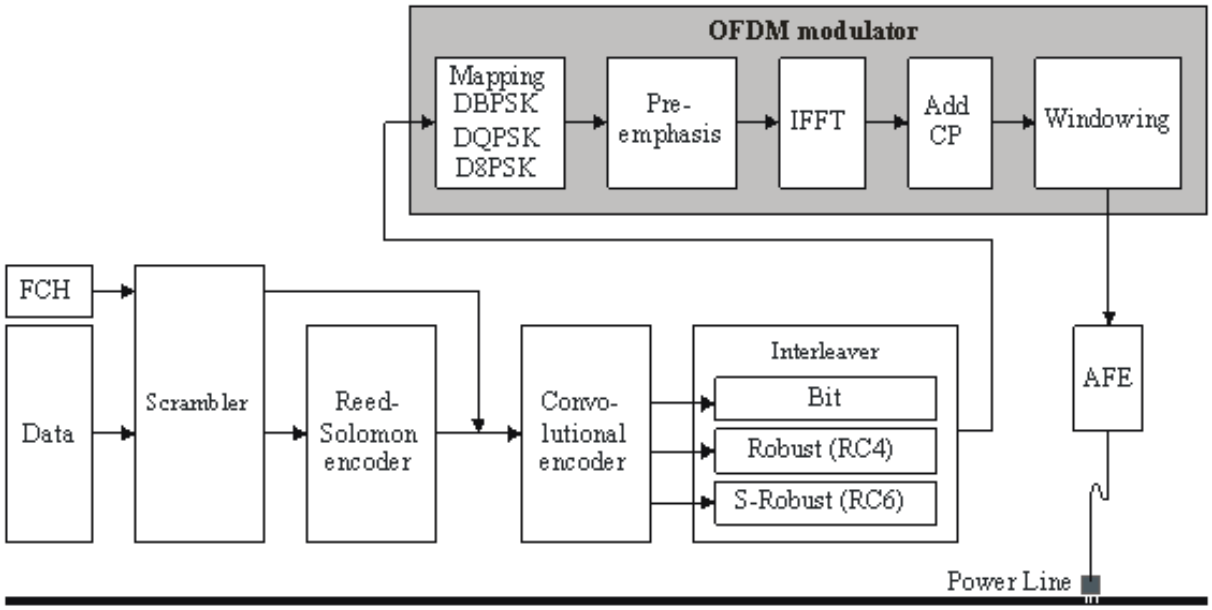


Figure 3: Block diagram of a G3-PLC OFDM transceiver

Each OFDM subcarrier uses one of the following differential modulations: **D8PSK, DQPSK and DBPSK**. Optional coherent modulations offering a higher level of robustness (BPSK, QPSK, 8PSK) may be used. **Robust mode**, consisting of a DBPSK modulation and a repetition code is a unique feature in NB-PLC, making G3-PLC particularly suitable for use in noisy environments. These modulations provide physical theoretical data rates of **up to 44 kbps in the CENELEC-A bandplan** and **up to 280 kbps in the FCC bandplan**.

In addition, **channel estimation allows G3-PLC neighbour nodes to continuously adapt the robustness/data rate compromise** by choosing the appropriate modulation but also to transmit data on the right OFDM subcarriers (“tone mapping”).

The abovementioned physical layer key features position G3-PLC as a technology suitable for harsh noisy environments typically observed on power lines.

From a data link layer perspective, the MAC layer is based on an adapted version of the original IEEE 802.15.4 [4] standard including an **enhanced CSMA/CA access method**, **acknowledgement of unicast frames**, **automatic repeat request (ARQ)** and **different priority levels to carry out QoS**. The IETF 6LoWPAN-based adaptation layer allows **IPv6 header compression** (to limit the impact of the overhead imposed by the IPv6 protocol) and **fragmentation** for optimal IPv6 support but also provides a G3-PLC node with a **routing algorithm optimized for the PLC medium: LOADng**.

Note: LOADng is implemented and enabled by default in all G3-PLC products. However, if required by the end-user, it is possible to disable the LOADng protocol[1].

Security is also present at several layers of the G3-PLC communication stack. **CCM* AES-128 ciphering** is provided by the IEEE 802.15.4-based MAC layer and **EAP-PSK authentication** is provided by the adaptation layer allowing the secure association of G3-PLC nodes to a G3-PLC network.

1.3. Application domains of G3-PLC

G3-PLC was originally developed for massive smart meter rollouts and, therefore, it was designed to deal with harsh channel conditions commonly found in the low voltage distribution network, by including features such as automatic adaptive modulation. G3-PLC is being used for smart metering globally, in different bandplans.

[1] Routing may be done at the IP level using LOADng or any other routing protocol.

Yet, as G3-PLC facilitates high-speed, highly reliable, long-range communication over existing powerlines, there are many more applications areas outside smart metering.

The main applications of the technology include smart metering, smart grid, street lighting, smart cities, railway control systems, building automation, home automation and further kinds of industrial applications where it makes sense to use the powerline as a communication medium.

Moreover, G3-PLC provides a high level of flexibility regarding network architectures. The protocol stack is modular, allowing for different options at different layers (see §1.1). G3-PLC may be used in a meshed configuration (typical architecture when deployed over a low voltage distribution network – see §3.4) or in a point-to-point configuration. Connectivity to remote devices, external to the G3-PLC network may be established in an end-to-end fashion or carried out through an applicative gateway (see §3.5). The location of different components such as the PAN-Coordinator (see §3.3), the authentication server (see §3.6), or the border router (see §3.5) may be located in different physical devices. Finally, native compatibility with the IPv6 protocol offers compatibility with a wide range of application layer protocols.

G3-PLC is constantly evolving to address new applications and challenges, making it successful in different countries!

2. The G3-PLC Alliance: standardization and certification

The G3-PLC Alliance is a non-profit association created in 2011 to standardize and promote the G3-PLC technology for smart metering, smart grids, smart appliances and industrial applications on a worldwide scale. The industry members of the G3-PLC Alliance (utilities, equipment manufacturers, chip manufacturers, test houses) are committed to its charter mainly consisting in the maintenance of the G3-PLC specifications [1] and their support in internationally recognized standards bodies, in facilitating interoperability among G3-PLC adopters and in promoting the G3-PLC technology.

Practically, the G3-PLC Alliance is organized in four workshops under the supervision of the Strategic Steering Committee (SSC):

- WS1: Specification Maintenance
- WS2: Marketing & Promotion
- WS3: Interoperability
- CPEC: Certification Program Maintenance

The involvement of the Alliance's members in the different workshops and standardization bodies led to the publication of the ITU-T G.9903 [3] standard. A liaison between ITU-T and the G3-PLC Alliance allows the maintenance work of WS1 to be directly injected into new corrigendum, amendments or revisions of the ITU-T G.9903 [3] standard.

Since 2014, the Alliance has started the operation of the G3-PLC certification program for the CENELEC-A, CENELEC-B, ARIB and FCC bandplans.

The publication of converged standards and the availability of a G3-PLC certification program positions G3-PLC as a mature solution for large scale rollouts of smart energy systems:

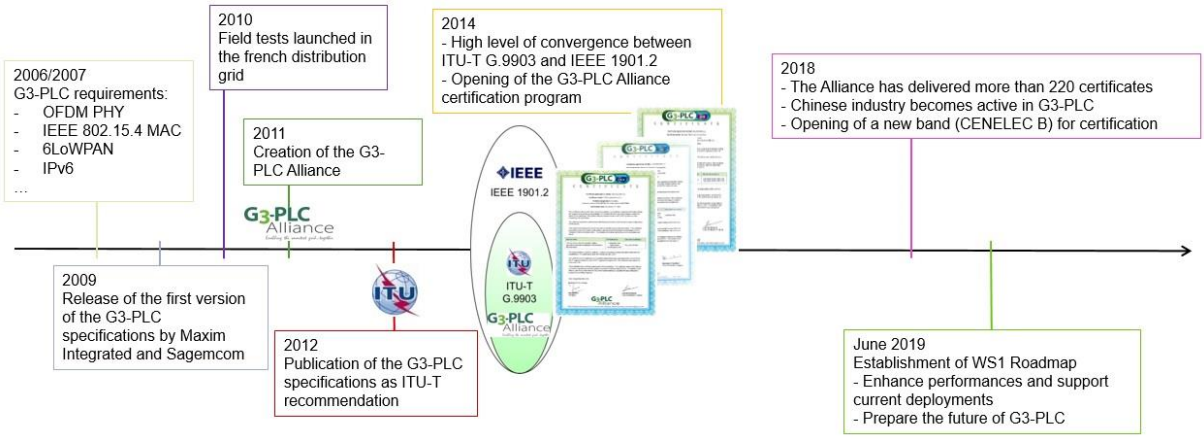


Figure 4: Standardization and certification milestones for the G3-PLC technology

The certification program of the G3-PLC Alliance is based on the three following fundamental pillars:

- **Conformance assessment**: aims to verify the conformance of the candidate implementation with the different clauses of the G3-PLC specifications [1].
- **Interoperability assessment**: sole conformance does not guarantee interoperability between different implementations. As a result, a candidate implementation is tested against a set of previously certified implementations.
- **Performance assessment**: performance tests are also necessary to prevent some issues to occur in the field. For instance, a too-long processing time may lead to application timer expiry and failure.

The G3-PLC Alliance certification program is operated by accredited test houses[2].

[2] An updated list of certified devices is available at the following link: https://g3-plc.com/g3-plc-certification/

3. Technical basics

3.1. Communication over the electrical network

3.1.1. General principles

Power Line Communication (PLC) consists in using electrical wires as a support for information transfer by adding a low power signal, usually between live and neutral wires. The superimposition of the PLC signal over an AC mains network is depicted in Figure 6:

- Time-domain: addition of low-amplitude high-frequency signal over the 50 Hz sinusoidal wave
- Frequency domain: presence of a group of low-amplitude kHz-range spectral lines in addition to the fundamental frequency of the 230 V / 50 Hz AC mains. Such a group of spectral lines spread over a band up to 350 kHz is typical of OFDM technologies such as G3-PLC.

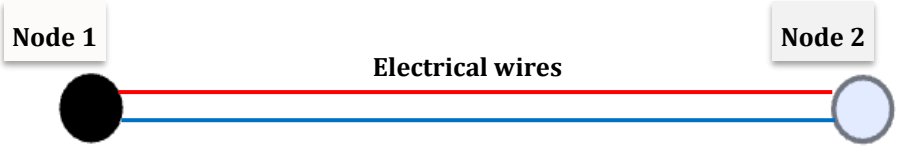


Figure 5: Two PLC nodes communicating over two electrical wires (live and neutral)

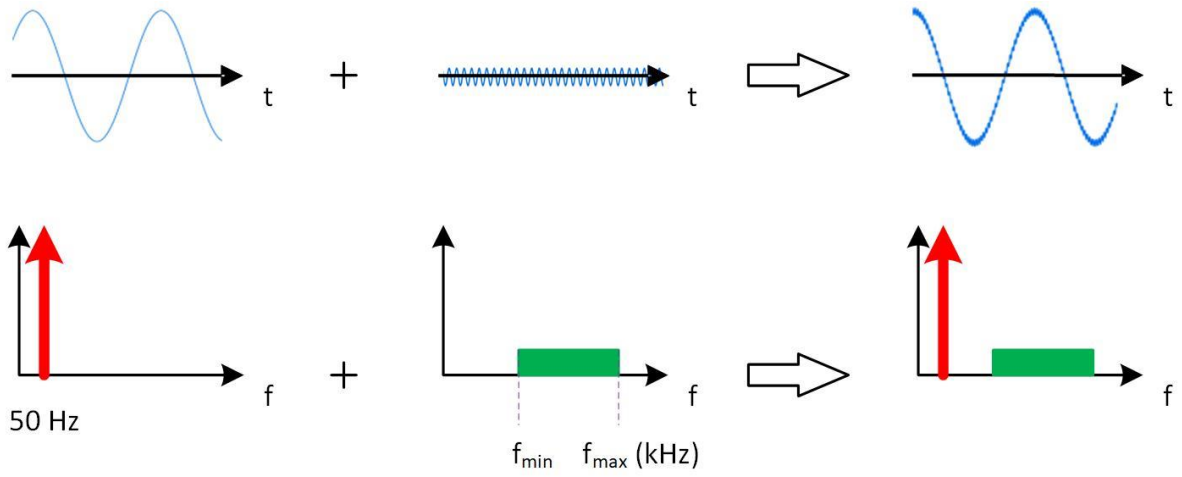


Figure 6: Representation of the superimposition of the G3-PLC signal over a 50 Hz AC mains network

Narrowband PLC devices (operating in the band 3 – 500 kHz according to international regulation as shown in §1.1) can be modelled as voltage generators, delivering a voltage which level has to stay as close as possible to a reference level under varying impedance conditions.

Signal levels are generally quantified as a voltage expressed in dBµV. Typical reference levels are in the range of 134 to 120 dBµV depending on the characteristics of the PLC technology implemented. The relationship between dBµV and Volts is illustrated in the following table:

Signal level in dBµV (logarithmic scale)	**Signal level in Volts** (linear scale)	
134	5	
126	2	
120	1	
114	0,5	
108	0,25	or 250 mV
88	0,025	or 25 mV
68	0,002 5	or 2,5 mV
48	0,000 25	or 250 µV

PLC modems are made of an analog stage (coupling, filtering and signal amplification) and a digital stage typically consisting of a Digital Signal Processor (DSP) and a Micro Controller Unit (MCU).

The coupler is making the interface between the modem and the electrical wires, by ensuring both galvanic insulation and a path for the high-frequency signal between the mains and the PLC device. When receiving a signal it is filtered, amplified and processed by the digital stage. Conversely, in transmission mode, these steps are executed in reverse order.

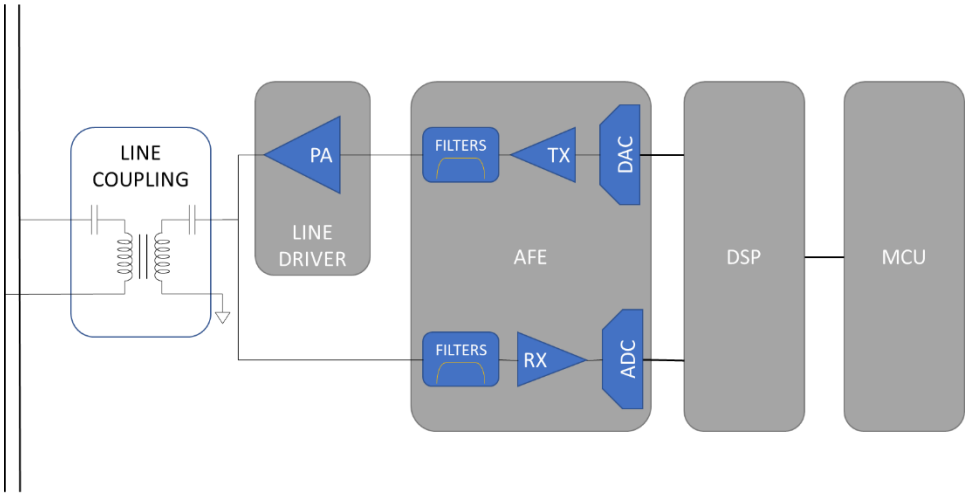


Figure 7: Schematic of a PLC modem

3.1.2. Environmental constraints and mitigation

The electrical network has not been designed for communication. Therefore, it is subject to impedance variations and the presence of noise in the operating frequency band of PLC technologies.

Impedance:

As a rule of thumb, network access impedance (i.e. the impedance measured between live and neutral at the location where a PLC device is physically connected) tends to decrease in dense urban areas, where a significant number of customers share the same electricity grid, compared to rural areas where long lines will only serve a few customers.

While the previous statement amounts to saying that, the more electrical branches, the smaller the impedance, impedance is also affected over time and varies with the number of appliances that are connected, switched on or switched off and their characteristics. For example, incandescent lamps modelled as purely resistive steady impedances are replaced by a growing number of LED lamps giving a more capacitive nature to the grid's impedance. Likewise, as it is observed with lighting equipment, appliances also embed more and more power electronics thus changing the nature of customer impedance.

Impedance cannot be known before the installation of PLC devices and is not constant over time. When impedance conditions become too challenging, power loss may be experienced by PLC devices (low impedances affect transmitters in particular) due to a mismatch between the grid impedance and the modem's internal impedance.

Attenuation:

Attenuation is described as the loss of signal between the PLC transmitter and the PLC receiver. It is a function of the distance travelled by the PLC signal and the frequency:

- The higher the operating frequency, the higher the attenuation.
- The longer the distance, the higher the attenuation.

Figure 8 shows how the low voltage distribution network (in red) has a low pass effect on the PLC signal resulting in attenuation. The attenuation, expressed in dB, also increases linearly with wire length.

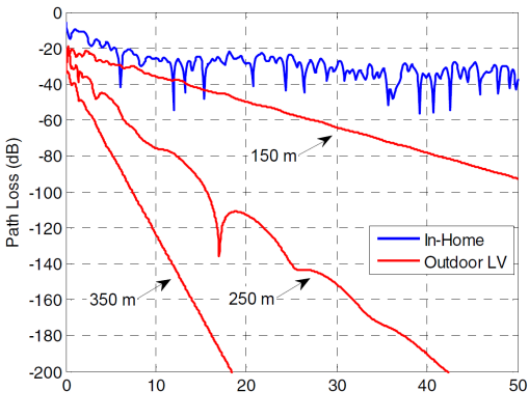


Figure 8: Attenuation in the MHz range for in-home and outdoor low voltage networks

Yet, Figure 8 essentially emphasizes the behaviour in the higher frequency range, up to 50 MHz, essentially used by broadband PLC technologies. If zooming in on the sub-MHz range, only variation of a few dB can be observed. In fact, in the sub-MHz frequency range, attenuation is more influenced by impedance mismatch issues created by the electrical network's topology, as described previously, rather than the losses strictly assignable to the propagation of the PLC signal.

As a conclusion, answering the question "what is the typical range of a narrowband PLC signal?" is almost impossible. It might be 600 meters in a rural area or 50 m in a very dense urban network.

Noise:

Another consequence of the shared nature of the electrical network is the possible presence of noise due to various electrical apparatus in operation. The growing pervasiveness of power electronics is one cause of the noise levels sometimes observed in the operating bands of narrowband PLC.

Practically, in addition to the attenuation effect due to impedance mismatch and signal propagation affecting the PLC signal level, noise is the second component entering in the determination of the signal-to-noise ratio (SNR).

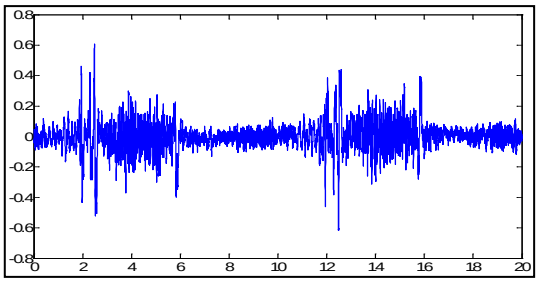


Figure 9: Example of a time-domain noise

The SNR is an essential metric in telecommunications, as it determines whether a receiver implementing a certain telecommunication technology will be able to decode the signal or not.

Mitigation techniques:

The PLC industry has developed hardware techniques and protocol mechanisms to meet the challenges of the electrical network as a propagation channel.

In idle and reception modes, G3-PLC modems are designed to present a high impedance to the grid (> 50 Ω).

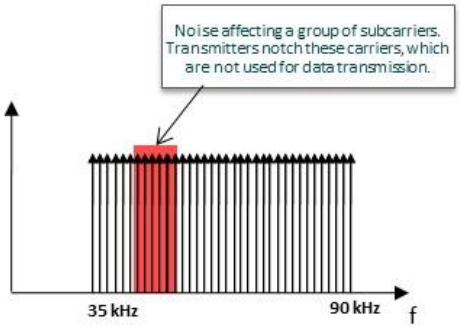


Figure 10: G3-PLC tone mapping

To deal with high attenuation levels, G3-PLC features dynamic adaptation of modulation schemes ranging from the most Robust modulation (DBPSK and repetition code) to modulations offering higher data rates (DBPSK, DQPSK and D8PSK).

G3-PLC may work in noisy environment thanks to the robust design of its physical layer, including a multi-symbol interleaver offering time diversity, and a retransmission scheme upon transmission failure detection.

In addition, G3-PLC takes advantage of frequency diversity offered by the OFDM modulation techniques by tone mapping consisting in avoiding transmitting useful information in the noisy spectral components of the operating band or compensating frequency-selective attenuation by adding a gain in specific frequency sub-bands.

Multi-phase propagation and crosstalk:

While the PLC signal is propagated along the wires to which the modem is connected, a portion of the transmitted signal is also found on the wires of a polyphase system. A signal initially injected between "Neutral" and "Live 1" may also be detected between "Neutral" and "Live 2" or "Neutral" and "Live 3" in a three-phase system. This physical phenomenon due to capacitive and inductive behaviour is called crosstalk.

Multi-phase propagation can be seen as a drawback, as it increases the "collision domain" (more PLC nodes can see each other creating more potential for congestion issues). But in general, multi-phase propagation helps the PLC signal to reach better coverage by offering more link diversity, thus increasing network density.

3.2. G3-PLC message structures

G3-PLC is designed according to the ISO Open Systems Interconnection (OSI) model. The specifications under the responsibility of the Alliance addresses the lower OSI layers up to the adaptation layer, generally defined as layer 2,5 (between data link layer 2 and network layer 3). Yet, the higher layer stack, which consists of the use of IPv6 at network layer and UDP as the preferable option for the transport layer, has to be considered with care.

Note: the use of TCP is also possible but sub-optimal with respect to the 6LoWPAN adaptation layer compression features. Therefore, TCP should only be used if required by the application protocol.

In line with the OSI model principles, the encapsulation and de-encapsulation processes of the G3-PLC are depicted in Figure 11.

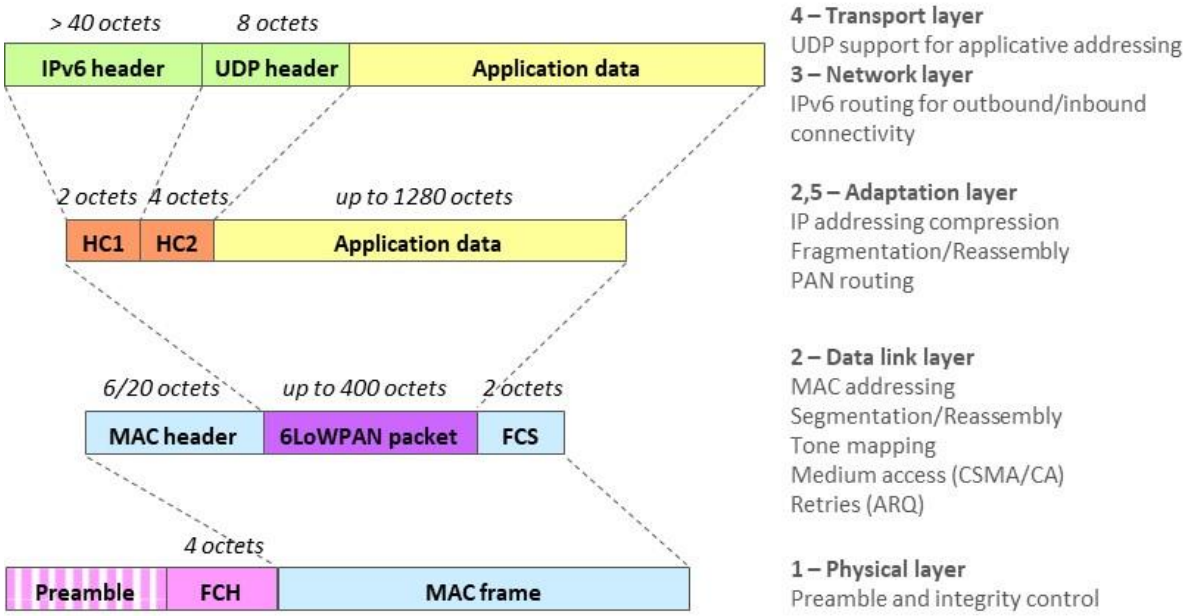


Figure 11: Encapsulation and de-encapsulation process principles (CENELEC-A bandplan)

As shown in the right part of Figure 11, different layers perform different actions with respect to their role in the OSI model. For example, the adaptation layer will compress IPv6 and UDP headers and, if applicable fragment the IPv6 packet into multiple 400-Byte 6LoWPAN packets.

In general, layer “n” receives data consisting of a header and a payload and processes this data according to the information carried in the header specific to layer n. For example, as shown in Figure 12, the MAC header contains information related to the destination address of the node this data is intended to; it may also include a request to re-negotiate the communication parameters (use of the “TMR” flag in the segment control field) with the node having sent this data.

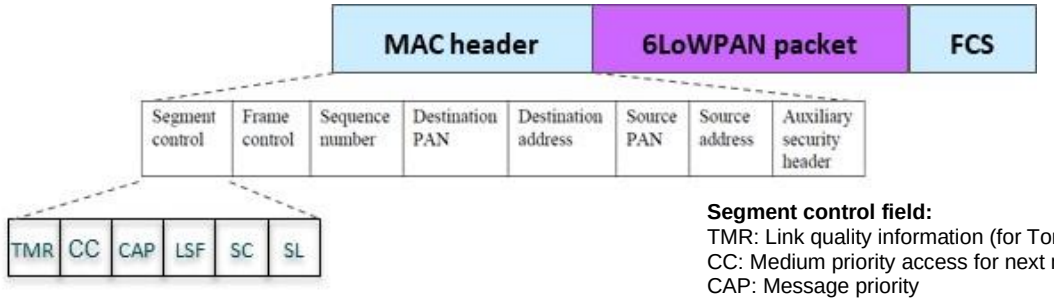


Figure 12: MAC header fields

3.3. G3-PLC network construction

On a three-phase network, G3-PLC nodes are connected to one or more phases and, most of the time, are transmitting/receiving PLC signals over only one phase. Using this physical infrastructure, the G3-PLC protocol builds a logical communication network (PAN; Personal Area Network, according to IEEE 802.15.4 [4] terminology), superimposed over the physical electrical network. Each PAN is typically identified by a 16-bit identifier (PAN-ID) statically assigned by the end-user. In general, each G3-PLC node is an application-independent device also known as “PAN-Device”. In addition, each PAN is managed by a supervisor node: the “PAN-Coordinator” grants access to the network to PAN-Devices after authentication and allocation of a 16-bit short address (unique in the PAN). The PAN-Coordinator typically provides connectivity to remote hosts through an external telecommunication network for every node.

Note: the use of a PAN-Coordinator is the general use case but it is possible to build networks using other application layer mechanisms, in particular for simple application scenarios.

One example of a G3-PLC PAN is the electricity metering infrastructure where PAN-Devices are smart meters and the PAN-Coordinator functions are completed by the data concentrator unit (which role mainly consists of collecting metering data).

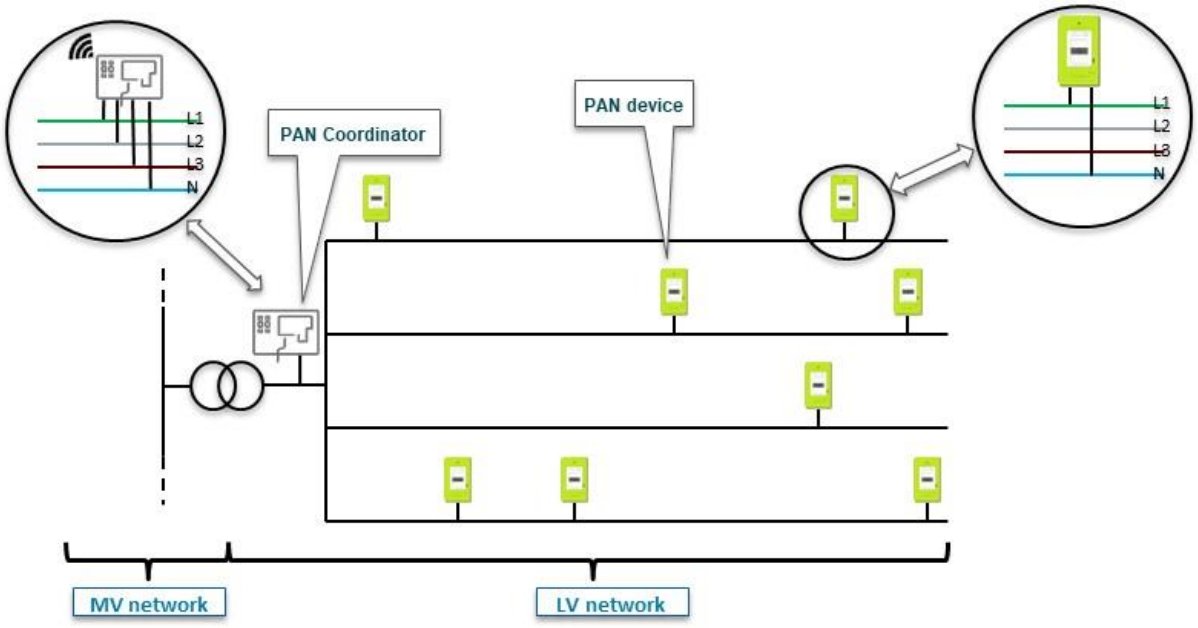


Figure 13: Example of a Linky G3-PLC PAN deployed over a low voltage distribution network

Moreover, the G3-PLC protocol is also designed and deployed to support various applications such as street lighting, smart cities or railway control systems as pointed out in §1.3.

With respect to other technologies with simple topologies (linear or star-shaped), G3-PLC allows the deployment of many nodes in complex environments. An optimal meshed, multi-hop network is built using existing link diversity between nodes: remote nodes are reachable using other nodes as relays as to establish end-to-end communications between any originator node and one (unicast communications) or multiple (multicast communications) destination nodes.

The mesh topology obtained is time-variant as links may vanish while others appear during the lifetime of the G3-PLC PAN (due to the change of network conditions as explained in §3.1.2). The G3-PLC routing protocol deals with these dynamic aspects as further explained in §3.4.

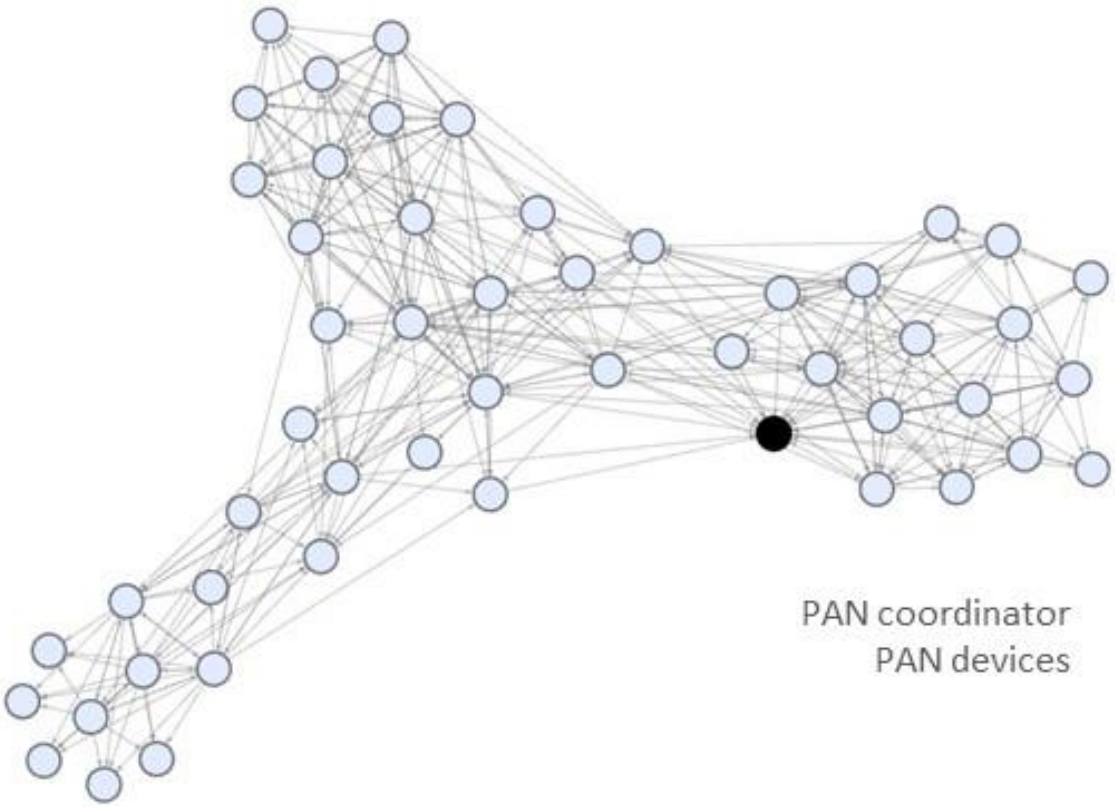


Figure 14: Example of a mesh topology of a G3-PLC PAN

Yet, before a mesh topology is established, all PAN-Devices typically have to carry out a secured association process following the specifications of the LoWPAN Bootstrap Protocol (LBP) defined in the G3-PLC specifications [1]. This specification is based on the well-defined EAP-PSK standard.

Before EAP-PSK authentication is carried out, a LoWPAN Bootstrap Device (LBD in LBP terminology) first explores his direct neighbourhood by sending a specific "beacon request" MAC command frame. Neighbours respond to this frame by sending a "beacon" MAC frame allowing the LBD to learn about his environment. If the LBD does not detect a good direct link to the LoWPAN Bootstrap Server (LBS), it selects a relay note also known as LoWPAN Bootstrap Agent (LBA) to which a good connection is found.

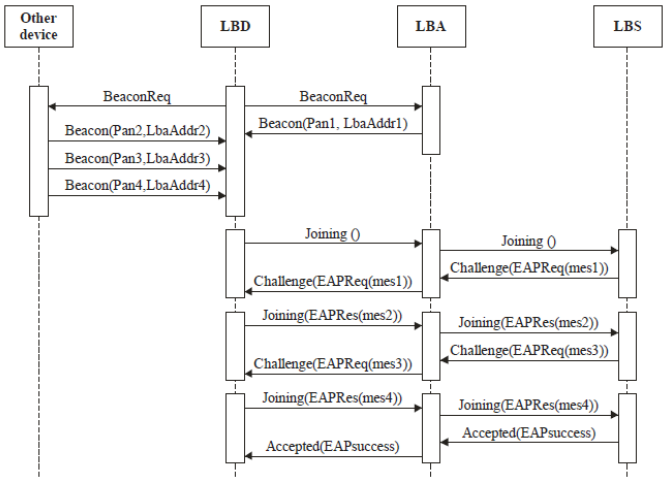


Figure 15: The LoWPAN Bootstrap Protocol (LBP)

The EAP-PSK authentication protocol is then carried out between LBD and LBS, using the selected LBA to forward information to/from the LBS. Both LBD and LBS need to share appropriate credentials before the LBS grants access to the G3-PLC PAN it maintains. Once the bootstrapping process has completed successfully, the LBD becomes an "associated" PAN-Device and is able to exchange information with any other node (PAN-Device or PAN-Coordinator) in the G3-PLC network.

3.4. Routing in a G3-PLC network

3.4.1. The LOADng routing protocol

The size of an electricity grid and its characteristics do not allow all nodes to transmit a message directly to the network coordinator. It is, therefore, necessary to use certain nodes as relays which are chosen to optimize transmission times and the use of available bandwidth. In addition, this choice can be questioned every time disturbances are encountered on the propagation channel (i.e. changes in impedance, attenuation or noise conditions as explained in §3.1.2).

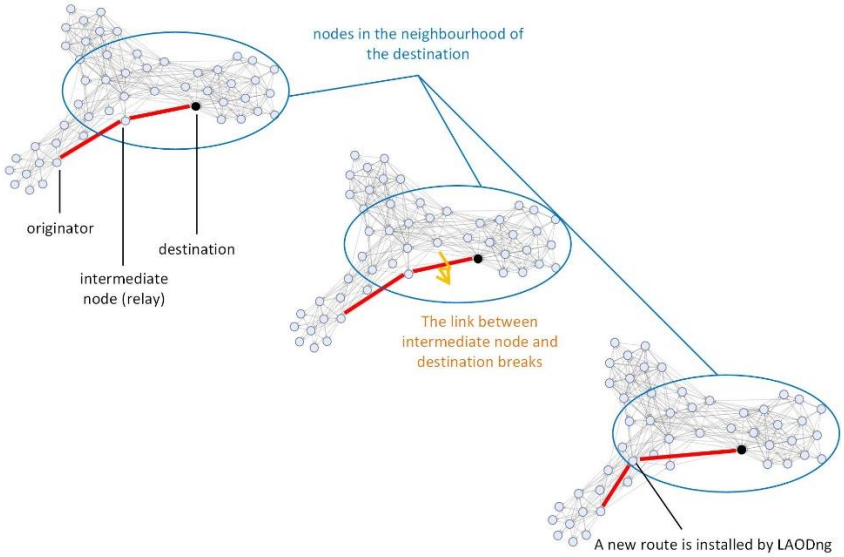


Figure 16: Restoring connectivity

Therefore, G3-PLC embeds a default reactive routing protocol, LOADng (Lightweight On-demand Ad-hoc Distance-vector routing protocol - Next Generation), responsible for finding the optimal path (according to a given routing metric) between an originator node and a destination node by selecting appropriate intermediate nodes acting as relays. LOADng allows each G3-PLC node to populate its routing table indicating the next hop towards a given destination. Furthermore, when a link breaks, as shown in Figure 16, LOADng allows to find an alternate path to restore connectivity without the involvement of upper layers.

LOADng is of reactive (as opposed to proactive) nature, as route discovery is triggered only if data needs to be transmitted, and if there is no available route to the destination node. In the G3-PLC stack, LOADng is implemented at data link layer (mesh-under) as it forwards 6LoWPAN frames using MAC addresses.

Basic principles are as follows:

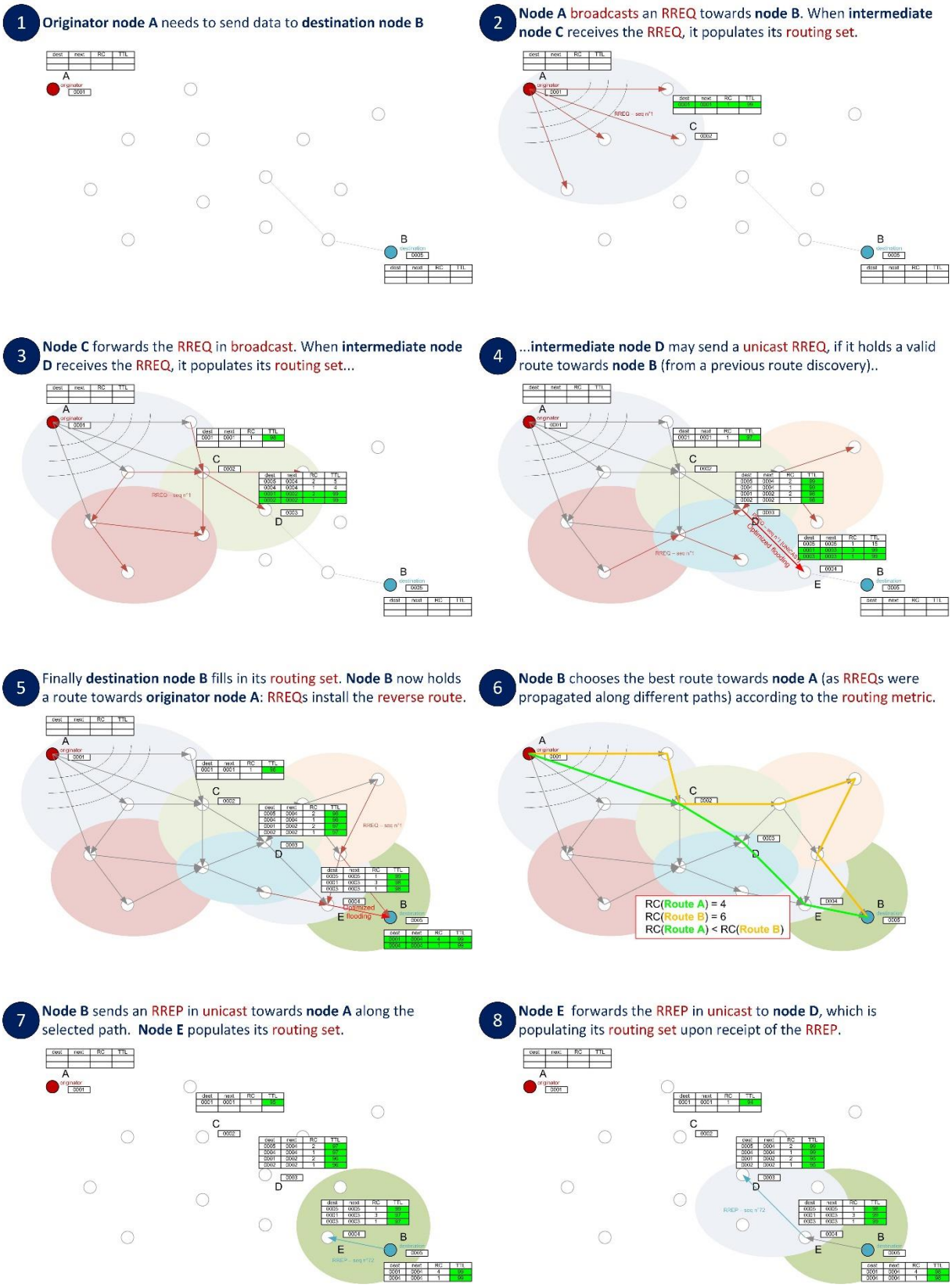
- Node A is about to transmit a packet to node B, and node B is not in node A's routing table.
- Node A initiates a route discovery by sending a Route REQuest (RREQ) message in broadcast, indicating that a route is sought to node B.
- While the RREQ is broadcasted towards the network from node to node, each node updates its routing set (temporary routing table entry), as depicted in Figure 17 (steps 2 to 5). The reverse route is being installed during the propagation of the RREQ message.
- Once node B has received several copies of the initial RREQ message having been propagated through different paths, it selects the best route according to the routing metric (Figure 17 – step 6) and sends a Route REPly (RREP) back to the originator of the RREQ (node A) in unicast along the selected path (Figure 17 – steps 7 to 10). The forward route is installed during the propagation of the RREP message.

In addition to these basic rules, LOADng allows to:

- Build bidirectional routes as link are often asymmetric (and even unidirectional) in G3-PLC networks. The bidirectional nature of a link is assessed by MAC acknowledgements during RREP propagation. The detection of a unidirectional link leads to the blacklisting of the node not having sent an acknowledgement.
- Reduce signalling traffic by allowing optimized flooding when the use of unicast RREQ is

enabled (Figure 17 – steps 4 and 5).

- Locally repair broken routes between intermediate nodes (instead of invalidating the whole route between originator and destination nodes).



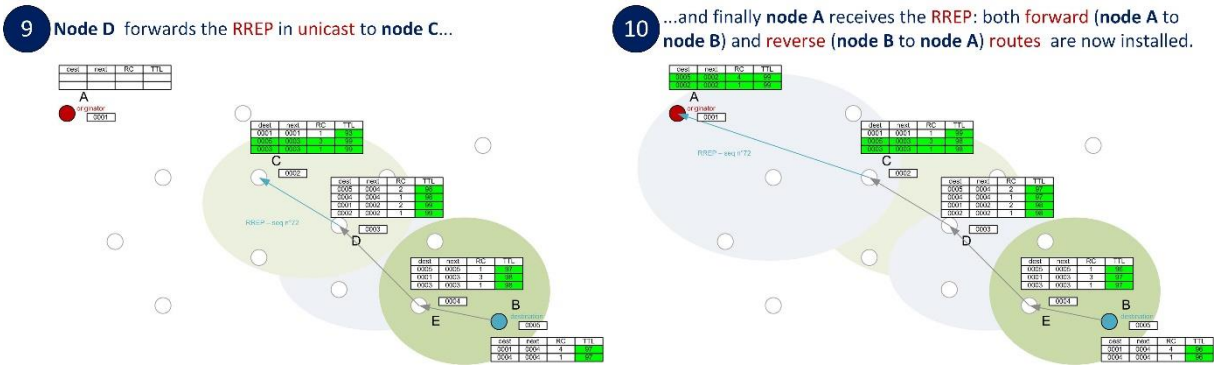


Figure 17: LOADng route discovery procedure

Once the LOADng protocol has installed a bidirectional route between an originator node and a destination node, the routing set entries installed by the different nodes along the selected path are converted into routing table entries for a given duration.

The local routing table attribute is maintained locally by each node and contains:

- The short address of the destination node
- The short address of the next hop (intermediate node) to the destination node
- The route cost along the path towards the destination node
- The hop count along the path towards the destination node
- The weak link count along the path towards the destination node
- A valid time

Note: weak links are links that are considered as fragile, due to a low signal to noise ratio (reflected in a low LQI). LOADng aims at minimizing the number of weak links in installed routes.

The validity of a route depends on the value of the time-to-live parameter (to be chosen according to the use case G3-PLC is addressing) and on the frequency of data traffic being forwarded along this route. Every time a frame is successfully propagated along a route to the destination, valid time is refreshed to its maximum value (time-to-live parameter) thus extending the validity of the route. On the other hand, if no data traffic is forwarded along the route or if a link breaks and local route repair was unsuccessful (a Route ERRor (RERR) is forwarded to the destination and the route valid time is set to 0), the route is no more valid.

Care shall also be taken when choosing a node's routing entry capacity; in a typical smart metering use case, it is recommended that a PAN-Coordinator can manage at least 1500 routing table entries, while 150 entries should suffice for PAN-Devices.

3.4.2. Route cost

Route cost is a metric that allows the determination of the best route, by adding up the link costs of each link along a path between an originator node and a destination node. LOADng leaves the choice of the best route to the destination node which receives RREQs having been forwarded along different paths, thus carrying different route costs.

In the example depicted in Figure 17, the metric chosen for computing the route cost is hop count, for simplification purposes. Indeed, in step 6, destination node B chooses the green "route A" (route cost = 4 hops) rather than the yellow "route B" (route cost = 6 hops) as it aims at minimizing the route cost.

In G3-PLC, the metric used to compute the route cost is a little more complicated than hop count, as indicators such as link quality or modulation type (a link using Robust mode will not provide any headroom in case a disturbance is encountered; therefore a link using D8PSK will be preferred) should also be taken into account.

G3-PLC specifications [1], define the “composite metric”:

$$LinkCost = max\left(C_{i\rightarrow j}, C_{j\rightarrow i}\right) + AdpKrt * \frac{NumberOfActiveRoutes}{MaximumNumberOfActiveRoutes} + adpKh$$

where $C_{i\rightarrow j}$ and $C_{j\rightarrow i}$ are the directional link costs (forward and reverse direction, respectively) between i to j. The directional link cost is computed as follow:

$$DirectionalLinkCost = adpKr * MOD_{Kr} + adpKm * MOD_{Km} + adpKc * \frac{(MaximumNumberOfTones - NumberOfActiveTones)}{MaximumNumberOfTones} + adpKq * MAX\left(0, MIN\left(1, \frac{adpHighLQIValue - LQI}{adpHighLQIValue - adpLowLQIValue}\right)\right)$$

The composite metric consists in taking into account the highest “directional link cost” as links are often of asymmetric nature as well as the number of routes already installed in a node’s routing table (with weighing factor “adpKrt”) and hop count (“adpKh”).

The directional link cost is a function introducing penalty when using Robust mode (weighed by “adpKr”) and depending on modulation type (weighed by “adpKm”), tone map (weighed by “adpKc”) and link quality (weighed by “adpKq”).

Note: changing the different weighing parameters to values different from the default values recommended in G3-PLC specifications [1] will change the way routes are established. Yet, if doing so, a thorough validation process has to be carried out before deploying the configuration at large scale.

3.5. Native support of IPv6

By design, the G3-PLC protocol stack allows native support of the IPv6 protocol, which grants end-user flexibility to fulfil business requirements when choosing the appropriate higher layers (ISO/OSI transport and application layers). This key feature also secures G3-PLC infrastructures in the long term, thanks to the scalability and future application compatibility provided by IPv6.

An IPv6 address is a 128-bit identifier (16 Bytes) that allows to uniquely identify a device in a network. It is made up of two parts:

- The 64-bit subnet prefix, which identifies the network the device belongs to.
- The 64-bit interface identifier, which identifies the device itself.

In a G3-PLC network, interface identifiers are always formatted as follows:

yyyy:00ff:fe00:xxxx (hexadecimal representation)

where **yyyy** corresponds to the PAN identifier (PAN-ID) and **xxxx** to the device’s short address (see §3.3).

Several types of IPv6 subnet prefixes are available for use in an IPv6 network:

- Link-local prefix: the support of this prefix is mandatory for any IPv6 device and is always available without configuration. Yet link-local addresses cannot offer end-to-end IPv6 connectivity from a host located outside of the local network.

  The link local prefix always equals:

  fe80:0000:0000:0000 (hexadecimal representation)

- Globally routable prefix (GUA): this prefix allows end-to-end IPv6 connectivity to/from G3-PLC PAN-Devices from/to hosts outside the local network, possibly on the Internet (these addresses are designed to be unique worldwide and can be used over the Internet).

  A globally routable prefix will be in the **2000::/3** (hexadecimal representation) address space, i.e. the three left-most bits are always equal to “001” (binary representation) while the remaining available 61 bits are allocated by a Regional Internet Registry (RIR) or an Internet Service Provider (ISP).

- ULA (Unique Local Address) prefix: this prefix allows end-to-end IPv6 connectivity to/from G3-PLC PAN-Devices from/to hosts outside the local network within an organisation but not from/to the Internet.

  A ULA prefix is formatted as a concatenation of the following fields:

  fc (7 bits – hexadecimal representation) + **L** (1 bit – equals 1 if locally assigned) + **Global ID** (40 bits, random value) + **Subnet ID** (16 bits, identifies a subnet within the organisation)

  Note: in practice organizations may use Global ID and Subnet ID fields as they wish to match their needs, and according to their internal addressing plans.

G3-PLC devices hold at least a link-local address and if end-to-end IPv6 connectivity is required outside the PAN, an organization may use GUA or ULA addresses.

To obtain a GUA or ULA address, a G3-PLC PAN-Device must be bootstrapped and already have a link-local address. 6LoWPAN “Neighbour Discovery” features allow the advertisement of the local GUA or ULA prefix of the PAN (PIO option in RFC 4861 [6]) and, ideally, additional context information (6CO in RFC 6775 [7]) as to enable external prefix compression for IPv6 packets originating from hosts located outside the local network.

Typically, the PAN-Coordinator becomes an IPv6 “border router” also known as “edge router” (even if the border router could be a dedicated PAN-device) establishing end-to-end IPV6 connectivity to/from the PAN directly from/to external hosts. Nevertheless, enabling IPv6 border router topologies (see example in Figure 18) goes together with reinforced security not limited to MAC ciphering and authentication as offered by G3-PLC: **cybersecurity best practices shall be implemented at system level in addition to what G3-PLC natively offers**.

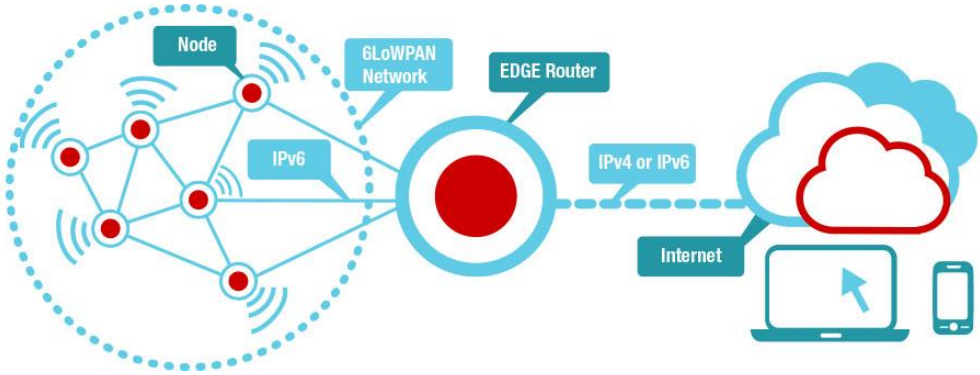


Figure 18: Example of a 6LoWPAN network connected to the Internet through an IPv6 border router

3.6. Cybersecurity aspects

As already introduced previously, cybersecurity is a key aspect to be considered in any network deployment including G3-PLC networks.

The G3-PLC protocol does implement state-of-the-art security as it specifies mechanisms ensuring mutual authentication and frame encryption at MAC layer (including data, routing or LBP configuration messages) as well as an anti-replay mechanism.

By design, security of G3-PLC PANs is typically managed centrally. The default EAP-PSK (Extensible Authentication Protocol – Pre-Shared Key) authentication method relies on a pre-shared secret: the PSK. Each PSK, specific to each PAN-Device, is known by both the PAN-Device itself and the authentication server either located in the central system or in the PAN-Coordinator.

3.6.1. Overview of the EAP-PSK authentication mechanism

The association of a new PAN-Device (LBD) is done in several steps (see §3.3). The PAN-Coordinator (LBS) is responsible for the authentication of LBDs before granting them access to the PAN by assigning short addresses and providing the shared communication key giving LBDs the ability to decrypt or encrypt data propagated in the PAN.

When a node carries out secured association according to the bootstrapping procedure introduced in §3.3, mutual authentication between this node and the authentication server is performed according to

the EAP-PSK protocol.

The LBP protocol (see §3.3) defines five types of messages:

"**JOINING**": request for association by any new node, carries embedded EAP-PSK messages during the authentication procedure.

"**CHALLENGE**": carries embedded EAP-PSK messages during the authentication procedure.

"**ACCEPTED**": carries embedded EAP-PSK messages in case of successful authentication, it is used to assign configuration parameters to the node (short address, GMK…).

"**DECLINE**": carries embedded EAP-PSK messages in case of authentication failure.

"**KICK**": allows the exclusion of an already associated node by the PAN-Coordinator.

At the end of the EAP-PSK procedure, if successful, the node is assigned a short address and is aware of the Group Master Key (GMK) used for MAC encryption: the node is now ready to securely exchange information on the G3-PLC network. The GMK is defined at the start-up by the PAN-Coordinator and is common to all the PAN-Devices of a same PAN.

3.6.2. Overview of MAC layer security: encryption and anti-replay mechanisms

Encryption is based on the use of the GMK previously shared by the PAN-Coordinator (after successful authentication) to all PAN-Devices. All messages sent by PAN-Devices are encrypted using this GMK and AES-128 algorithm with CCM* mode of operation. MAC frame payloads are encrypted, together with security headers used for decryption by recipients.

The CCM* mode requires the definition of an initialization vector to be used as an input parameter for encryption and ensures that encrypted frames are not played back.

MAC encryption generates a slight increase in the size of transmissions:

- The MAC header of an encrypted frame contains a 6-Byte security header consisting of 3 fields:
    - Security level (1 Byte): security level of the frame
    - Key identifier (1 Byte): GMK key identifier
    - Frame counter (4 Bytes): used for the initialization vector and the anti-replay mechanism
- A 4-Byte "Message Integrity Code" (MIC) field is added after the encrypted data. The MIC provides protection against any alteration of the content of the frame or its header.

According to common recommendations, the PAN-Coordinator has the ability to modify the GMK used by all PAN-Devices thanks to a dedicated "re-keying" process.

4. SLA (Service Level Agreement) expectations

Like many other wired or wireless telecommunication standards, the G3-PLC protocol stack is not designed for deterministic networking and does not provide resource allocation mechanisms. Therefore, performances may vary significantly if higher application layers mechanisms are not appropriately chosen: a G3-PLC channel can obviously not be used the same way as a high throughput medium.

That being said, this section provides reference data on what can be expected in a G3-PLC network composed of devices with correct analog design (parameters such as output level, in particular in low-impedance conditions, shall not be neglected) and possibly embedding additional digital processing such as blanking or clipping techniques, out of scope of the G3-PLC specifications [1].

Dynamic range:

The dynamic range is defined as the maximum attenuation a PLC signal can undergo before the PLC link between two nodes breaks. In practice, the dynamic range corresponds to the attenuation measured between a transmitter and a receiver (device under test) when the Frame Error Rate (FER) reaches a value of 5%. The dynamic range can be assessed in ideal conditions (no noise, moderate impedance), under low impedance conditions and/or in the presence of noise.

A certified implementation guarantees a dynamic range of at least 60 dB without noise and in moderate impedance conditions (such as those provided by a CISPR 16-1-2 [8] Artificial Mains Network – AMN), and as a rule of thumb a value higher than 80 dB is often measured. It is more difficult to provide reference values in the presence of noise as implementation choices can be differently affected by the same noise waveform.

The dynamic range may be assessed using the methodology described in ETSI TS 103 909 [9] (note that this technical specification uses the term link budget).

Theoretical physical layer data rates, point-to-point UDP throughput and point-to-point latency:

The theoretical physical data rate corresponds to the maximum data rate that can be achieved at physical layer in ideal conditions, and thus, it does not take into account channel access times or possible disturbances observed in the propagation channel. Yet, it is the main data rate indicator used by all telecommunication technologies. G3-PLC can offer data rates up to 44 kbps in CENELEC-A bandplan and up to 280 kbps in FCC bandplan.

Another useful indicator is the UDP throughput which can be achieved between two directly connected G3-PLC nodes, as it reflects the real data rate available for the application layer. The UDP throughput has been established in the laboratory using dedicated well-known tools (iperf or jperf): 13 to 17 kbps have been achieved in CENELEC-A bandplan and 66 to 83 kbps have been reached in the FCC bandplan.

Finally, point-to-point latency has been established in the lab by measuring the ICMP ping roundtrip time between two directly connected G3-PLC nodes. 120 ms have been measured in the CENELEC-A bandplan while a value of 56 ms has been established in the FCC bandplan (64-Byte payload size).

The following table summarizes the abovementioned results:

	Theoretical PHY data rate	**Typical UDP throughput**	**Typical ICMP ping roundtrip time**
CENELEC-A	Up to 44 kbps	13 to 17 kbps	120 ms
CENELEC-B	Up to 19 kbps	5 to 7 kbps	210 ms
FCC	Up to 280 kbps	66 to 83 kbps	56 ms

- **Theoretical PHY data rates of any communication technology should be interpreted with care!**

Theoretical PHY data rates result from a mathematical computation taking into account PHY parameters such as the structure of a PHY frame, the characteristics of the modulation used (OFDM for G3-PLC: number of Fast Fourier Transform points, number of subcarriers, etc.), the sampling frequency, error correction coding, etc. Paragraph 7.3 of the G3-PLC specifications [1] gives an example of such computation for different bandplans of the G3-PLC technology.

In the field, a communication technology's performance is influenced by many factors. Some,

such as noise or interferences, may impair its operation. In addition, intrinsic mechanisms of the protocol stack, such as channel access or routing (for mesh networks) may also impact performance as they influence the timing at which messages are exchanged.

Hence, **the computed PHY data rate is a pure theoretical indication:** some technologies favour higher data rates at the cost of lower robustness (a PHY frame of a given length will include less error correction coding and more payload) while others favour robustness over speed (a PHY frame of a given length will include more error correction coding and less payload). Of course, less robustness increases transmission errors and retries, which will result in loss of time and a reduced data rate (the same quantity of data is sent over a longer period than expected).

In conclusion, an end-user shall always consider typical throughput obtained just below the application layer (for example the UDP throughput in this document) and latency instead of focussing on theoretical PHY data rates. In addition, empirical experience with the technology and performances reported for similar application use cases shall be taken into account when selecting a communication technology.

Empirical high-level SLA for the smart metering use case:

Performances in the field vary according to environmental factors such as the number of nodes in a G3-PLC network, the density of nodes and their geographical distribution, presence of noise, impedances, attenuation and traffic patterns. Due to the meshed nature of G3-PLC PANs, the rollout strategy (massive with saturated PANs or occasional over several years) will also have an impact on the performances during the deployment of the smart metering infrastructure.

Experience shows that G3-PLC allows average daily collection rates of 98% on large scale deployments (more than 1 million units installed over several PANs), thanks to the robustness of the technology as well as appropriate application layer policies.

It is also well-assumed that a medium-sized smart meter network consisting of up to 200 nodes could be read out within 30 minutes. Reaching one individual node (without parallel requests to other nodes) in a multi-hop G3-PLC network could take less than 1 second if there is a valid route to this node and up to a few seconds if a new route has to be built.

5. Abbreviations

6CO	6LoWPAN Context Option
6LoWPAN	IPv6 Low power Wireless Personal Area Network
AES	Advanced Encryption Standard
ARIB	Association of Radio Industries and Businesses
ARQ	Automatic Repeat request
CENELEC	European Committee for Electrotechnical Standardization
CISPR	International Special Committee on Radio Interference
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance
D8PSK	Differential 8 Phase Shift Keying
DBPSK	Differential Binary Phase Shift Keying
DQPSK	Differential Quadrature Phase Shift Keying
DSP	Digital Signal Processor
EAP	Extensible Authentication Protocol
FCC	Federal Communications Commission
FER	Frame Error Rate
GMK	Group Master Key
GUA	Global Unicast Address
ICMP	Internet Control Message Protocol
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IP	Internet Protocol
ISO	International Standards Organization
ITU	International Telecommunication Union
LBA	LoWPAN Bootstrapping Agent
LBD	LoWPAN Bootstrapping Device
LBP	LoWPAN Bootstrapping Protocol
LBS	LoWPAN Bootstrapping Server
LOADng	6LoWPAN Ad-hoc On-demand Distance vector routing – Next Generation
LQI	Link Quality Indicator
MAC	Medium Access Control
MIC	Message Integrity Code
MCU	Micro Controller Unit
NB-PLC	NarrowBand Power Line Communication
OFDM	Orthogonal Frequency Division Multiplexing
OSI	Open Systems Interconnection
PAN	Personal Area Network
PIO	Prefix Information Option
PLC	Power Line Communications

PSK	Pre-Shared Key
QoS	Quality of Service
RERR	Route ERRor
RREP	Route REPly
RREQ	Route REQuest
SLA	Service Level Agreement
SNR	Signal to Noise Ratio
ULA	Unique Local Address
UDP	User Datagram Protocol

6. References

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